DeepReservoir: Deep Reinforcement Learning (DRL) for Optimal Control of Water Reservoirs

Shubhendu Kumar Singh, Cristina Garcia Cardona, Nishath Rajiv Ranasinghe, Susan Behery, Katrina E Bennett, Jon Schwenk

Freshwater demand continues to grow, propelled by rapid population growth, expanding agricultural activities, and future unprecedented climates that all act to further stress existing water resources. In the Southwestern USA in particular, historic water rights agreements are becoming more challenging to maintain as these compounding pressures mount. To adapt to these growing demands, changes in hydrology, and existing compacts that must be honored, we explore the use of deep reinforcement learning (DRL) as a tool to aid the management of water reservoirs. This approach features an "agent," analogous to a reservoir operator, which is a neural network-based algorithm, that interacts with a simulated environment to develop optimal reservoir management strategies. The framework is adaptable to multiple objectives and varying meteorologies while meeting multiple goals and considering forecasts. At the crux of this approach is the DRL agent, the proximal policy optimization (PPO) agent, which makes decisions to optimize reservoir operations. In our work, we consider the Navajo Dam in Northern New Mexico as a test case. We developed an LSTM-based inflow model spanning the four major gaged catchments upstream, as well as one ungauged basin. We also trained an LSTM to represent evaporation from the reservoir. Thus, the physical environment is completely defined by AI models driven by external meteorological drivers. To date, we have found that the DRL system can readily meet fairly simple objectives with a degree of robustness. We also subject the trained system to a set of “out of sample” meteorologic conditions to assess its adaptability to possible hydrologic changes. Our goal is to gradually increase the complexity of the constraints to approach the reality of the demands of reservoir management. Our initial results showcase the efficacy of the proposed framework while highlighting its ability to adapt to evolving real-time conditions, bringing us closer to providing a more data-driven and adaptive tool to help inform reservoir managers striving for optimality

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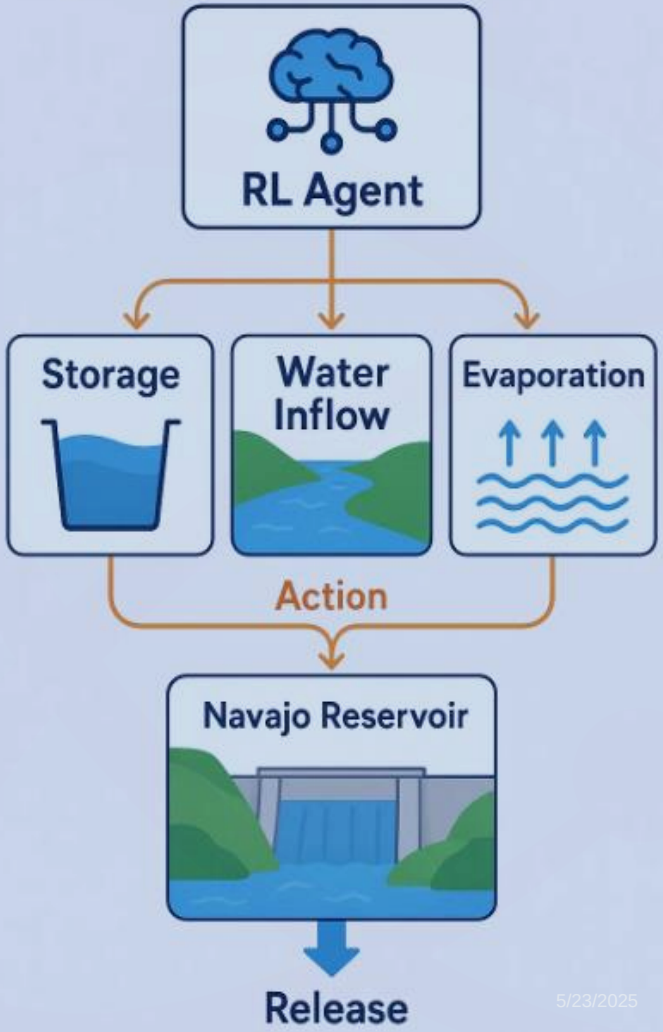

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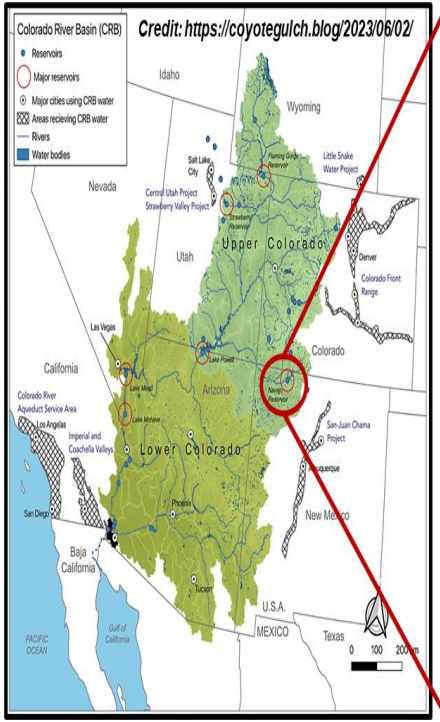
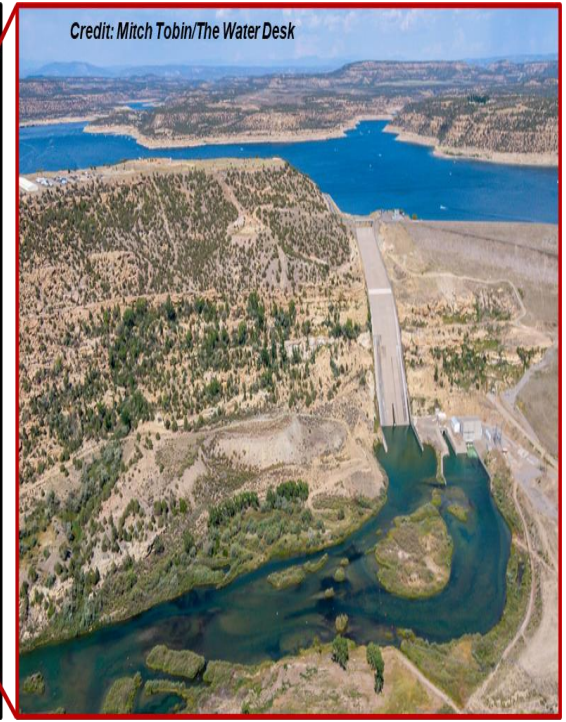
Objective

- **Critical Regional Asset:** Supplies water, irrigation (San Juan Basin), flood control, hydropower, and recreation across NM & CO.
- **Manual Rule-Based Decisions:** Operators follow pre-defined rule curves, which lack flexibility during extreme events.
- **Uncertainty in Hydrologic Conditions:** Inflow is influenced by snowmelt, rainfall, and upstream conditions — making precise planning difficult.
- **Balancing Conflicting Demands:** Must simultaneously manage flood control, water supply, ecology, and recreation — often with trade-offs.
- **Risk of Sub-Optimal Operations:** Static or reactive decisions can lead to overflow, low storage, or failure to meet downstream needs.





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Is reinforcement learning a viable tool for delivering realistic, data-guided, optimal solutions and exploring alternative management strategies?

Why Reinforcement Learning??

🧠 **RL enables adaptive control**: Learns optimal release strategies through interaction, improving resilience and efficiency.

🌐 **Scalable framework**: RL can generalize to similar reservoir systems under changing climate and hydrologic conditions.

✅ **Proven effective:** RL has been crucial to a number of cutting-edge technologies

Deep Reinforcement Learning (DRL) for Reservoir Control

🌍 **Virtual Environment**

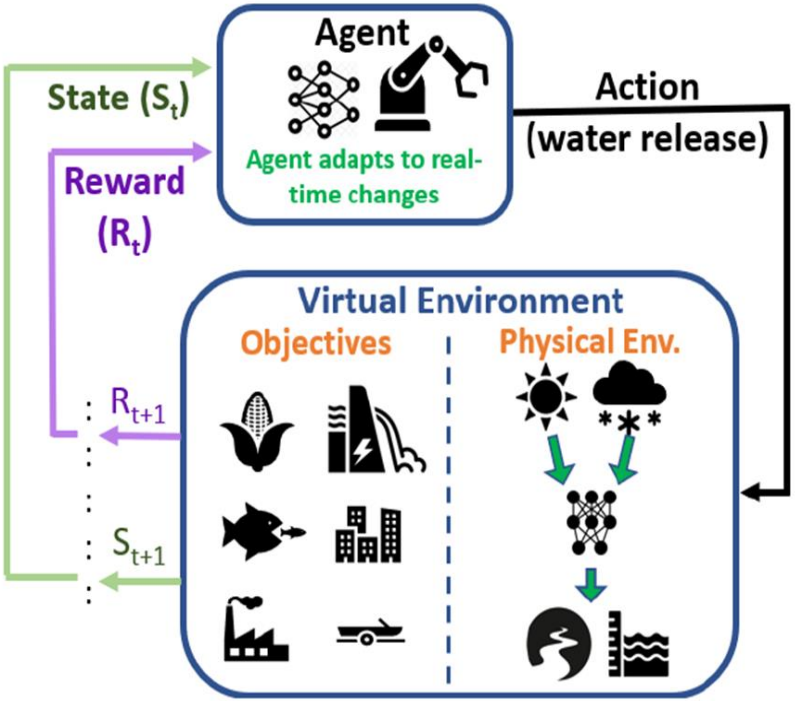
- Simulates water release management for Navajo Reservoir
- Captures reservoir dynamics: inflow, storage, evaporation, and releases
- Includes the management objectives

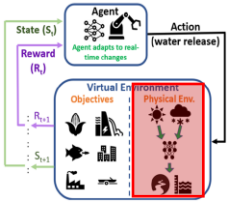
📊 **State (Observation Space)**

- **Storage**: Current volume of water in the reservoir
- **Evaporation**: Estimated water loss due to evaporation
- **Inflow**: Daily inflow into the reservoir

🤖 **Agent (Actor - PPO)**

- **Action**: Determines daily water release based on observed state
- **Objective**: Maximize efficiency while satisfying operational constraints
- **Algorithm**: *Proximal Policy Optimization (PPO)* for stable learning



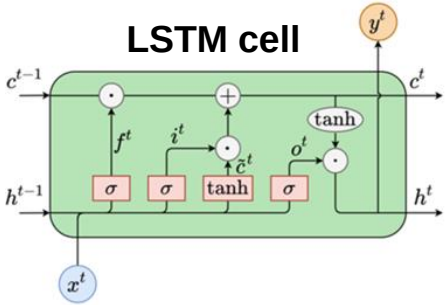


Physical Environment- Streamflow Model

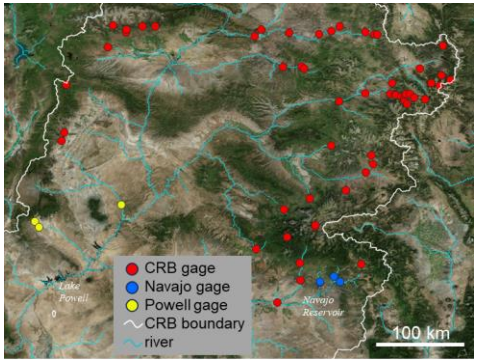
- **LSTM** model to estimate streamflow was trained with the following data:
  - *precipitation, wind speed, max temp., min temp.*
  - *snow water equivalent, soil moisture*

- **LSTM** models outperform process-based models in modeling streamflow in ungauged basins utilizing remote sensed data

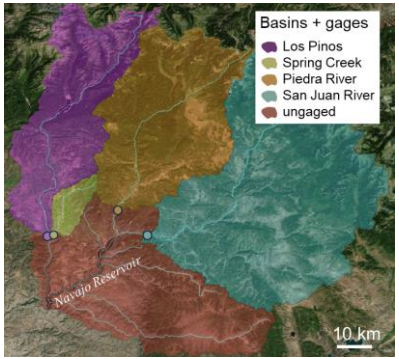
LSTM cell



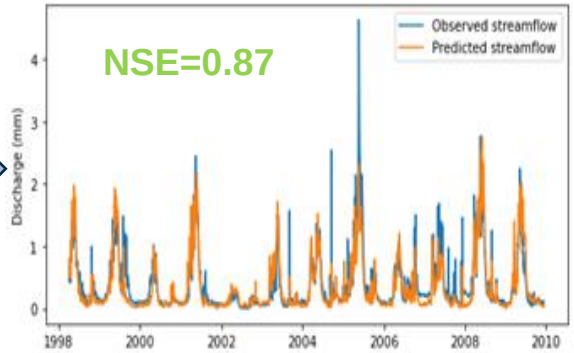
Training



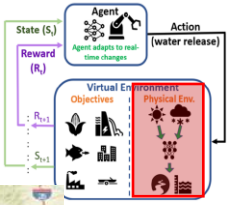
Fine Tuning



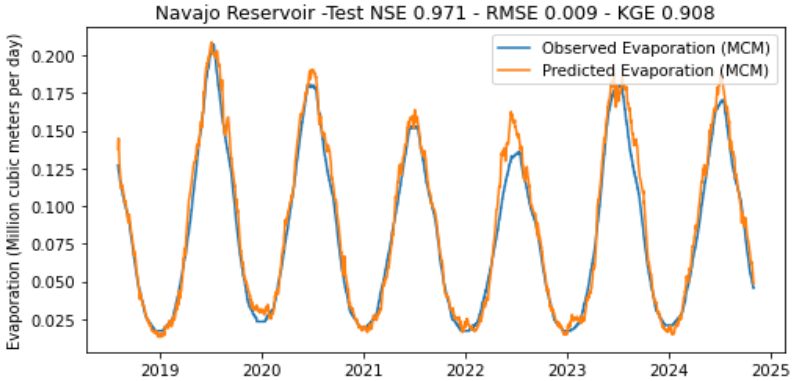
Testing- Reservoir Inflow



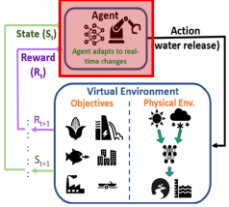
Physical Environment- Evaporation Model



- LSTM model to estimate evaporation was trained using 11 nearby reservoirs with the following data:
  - In-situ measurements – *storage volume, elevation, total release, input flow*
  - ERA5 meteorological inputs – *median temperature, mean wind speed, total precipitation*

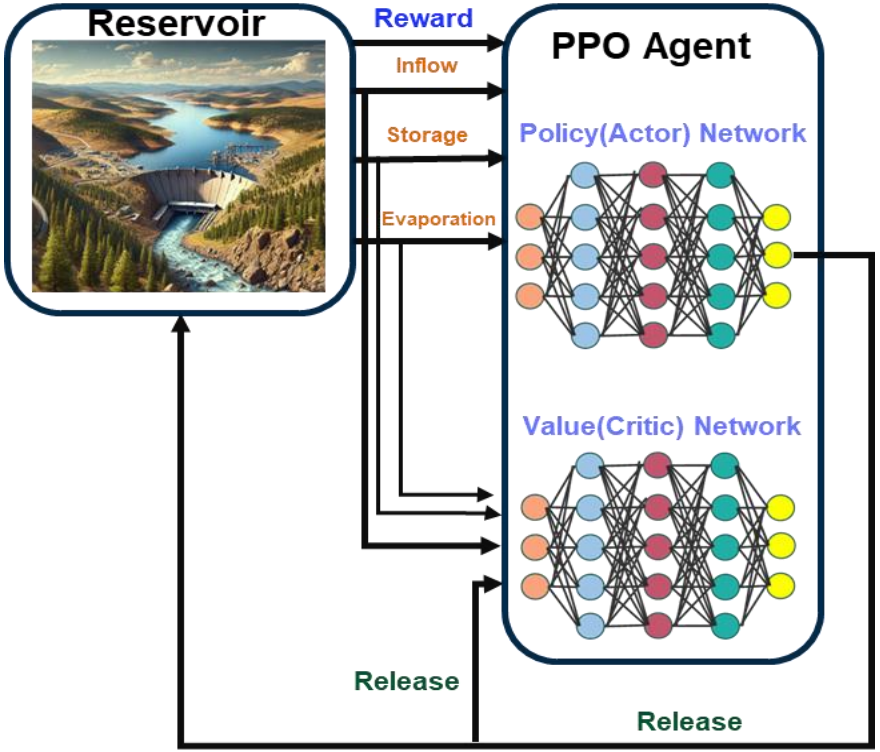


Agent - Proximal Policy Optimization (PPO)

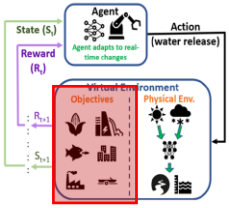


Why PPO for Navajo Reservoir?

- Handles **continuous action space**: realistic release decisions
- Takes all relevant inputs (inflow, storage, evaporation, forecasts) to provide adaptive and **data-driven control**
- Learns **anticipatory release patterns**, balancing short-term demands and long-term reservoir health



Reward Functions to Achieve Objectives

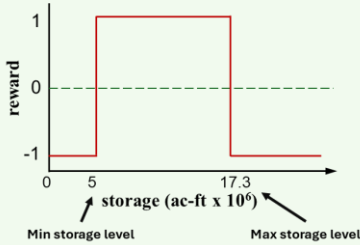


How to best-design reward functions in an RL context?

Objective:
Release water while maintain the storage with min and max storage limits

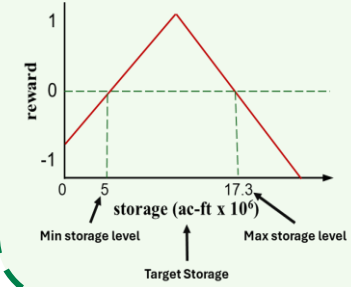
simplest approach

Reward Function:
+1 if the storage within the desired range else -1



more reward for being near mid-storage

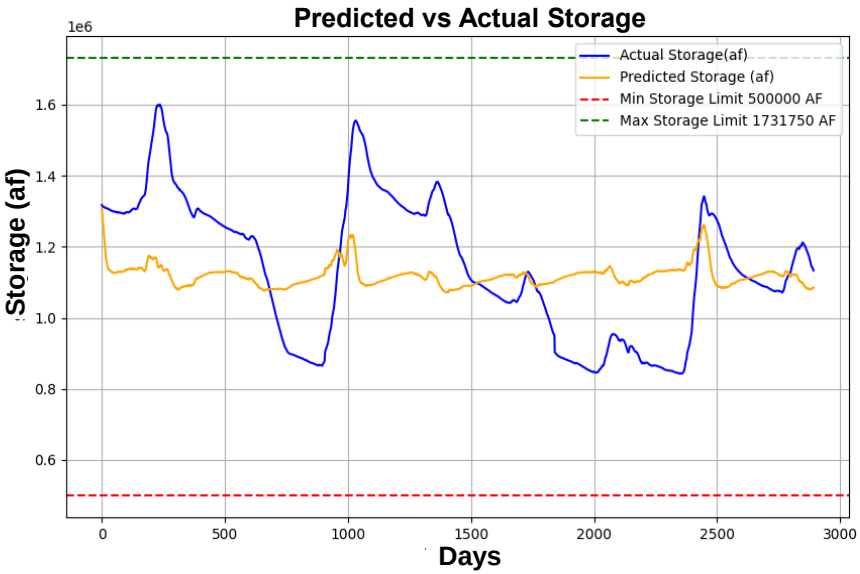
Reward Function:
+1 if the storage equal to target else less than 1

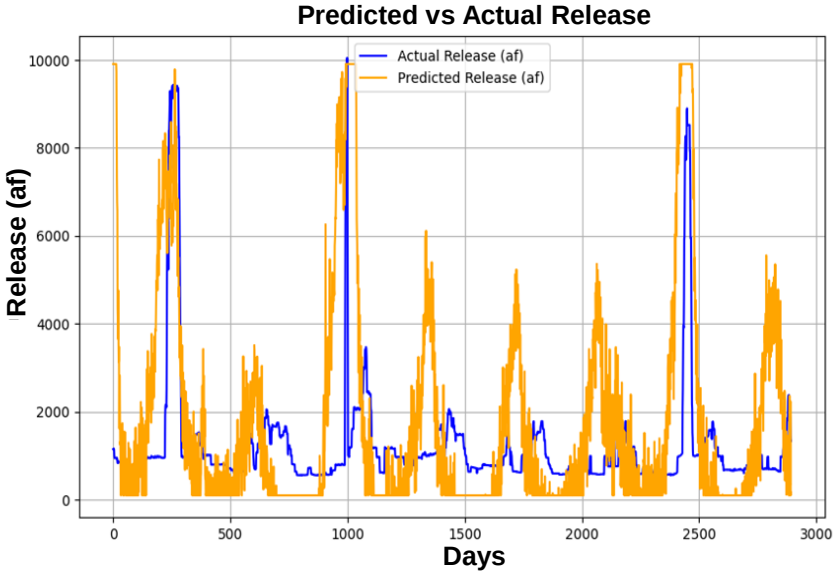


A Simple Objective

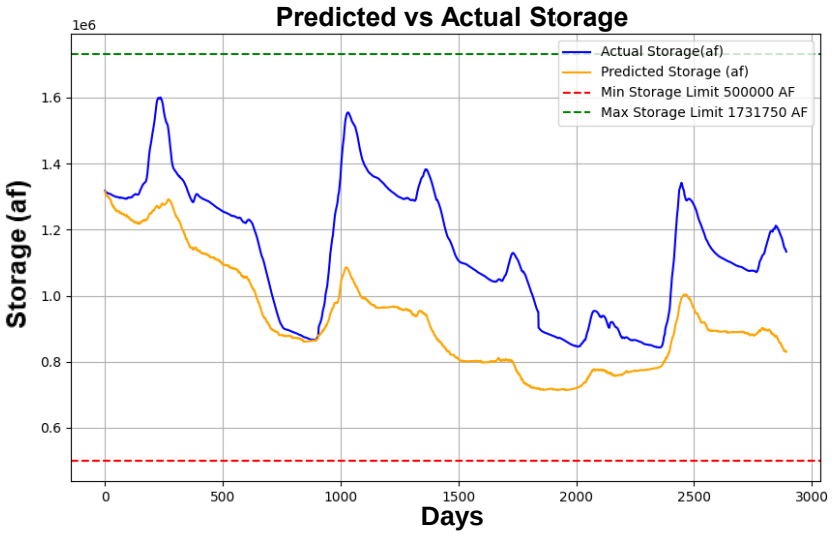
Reward :

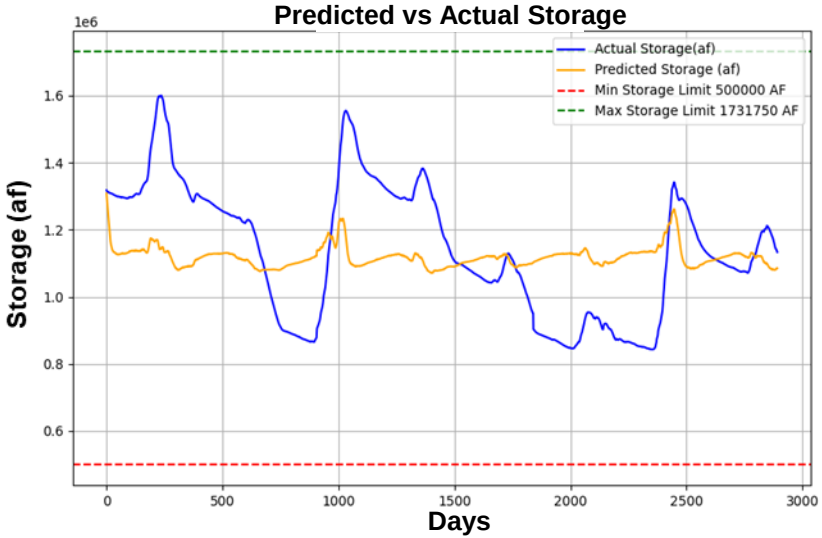
- Min-max limits with maximum reward for the midpoint of the limits





Effect of Reward Function Design





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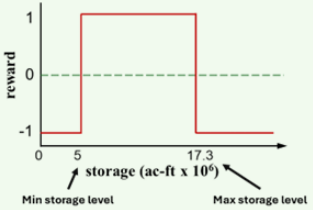

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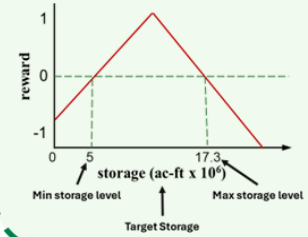
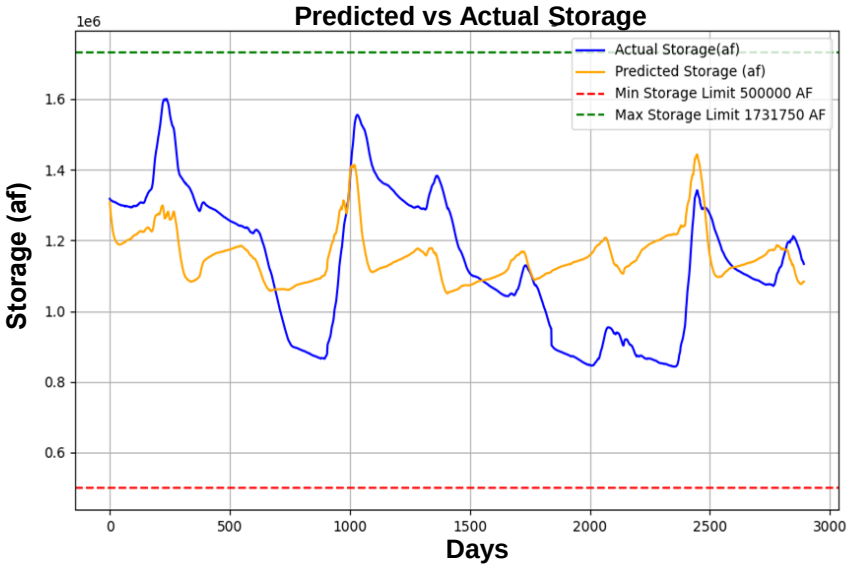
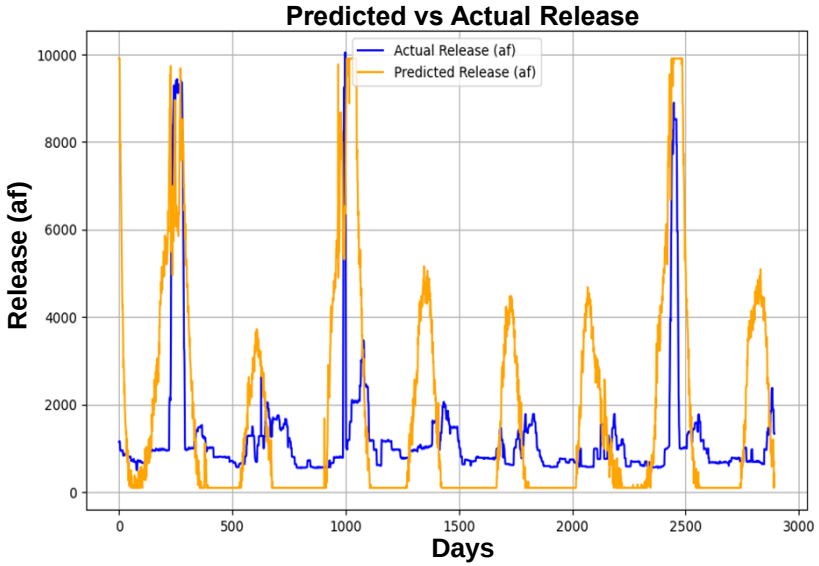


Enforcing Smoothness

Reward :

- Min-max limits with maximum reward for the midpoint of the limits
- Penalty for abrupt release

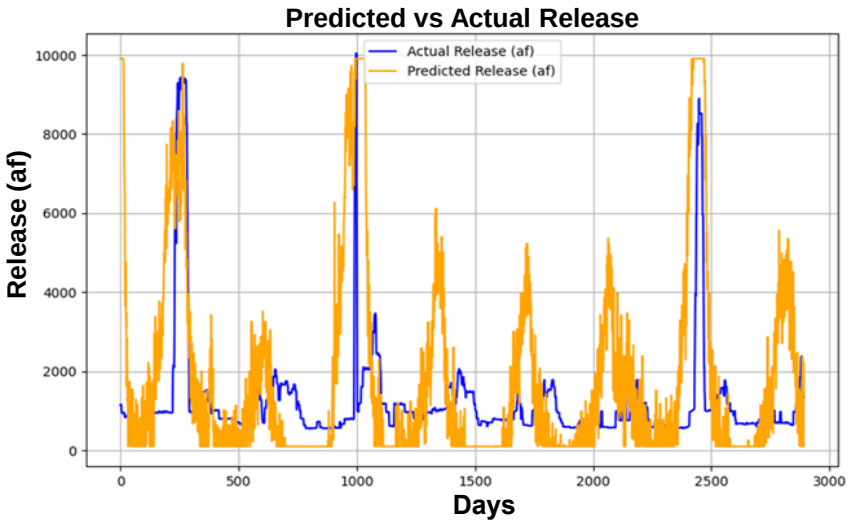




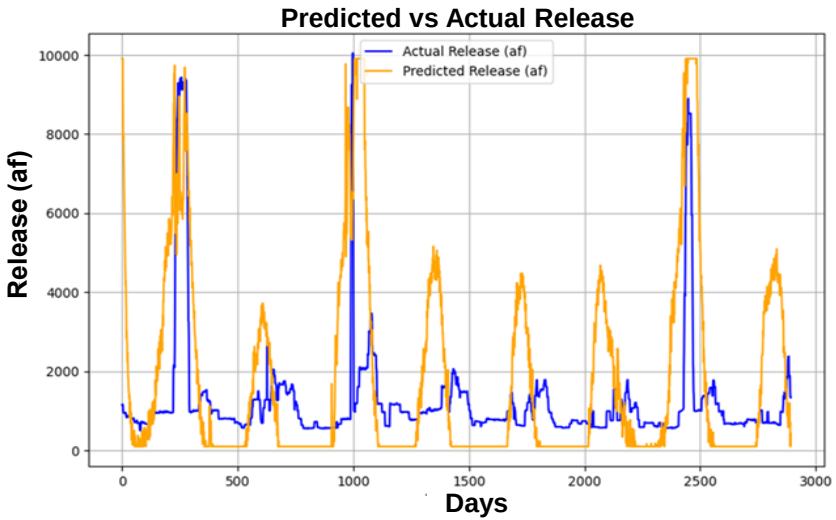
Quantifying Smoothness

$$\textit{Smoothness Metric} = 1 - \frac{\sum_{k=kc}^{N-1} P_k}{\sum_{k=0}^{N-1} P_k}$$

Where:
P_k represents the energy at each frequency component
K_c is the cut-off frequency index
S ε [0.1], where higher S indicates smoother release



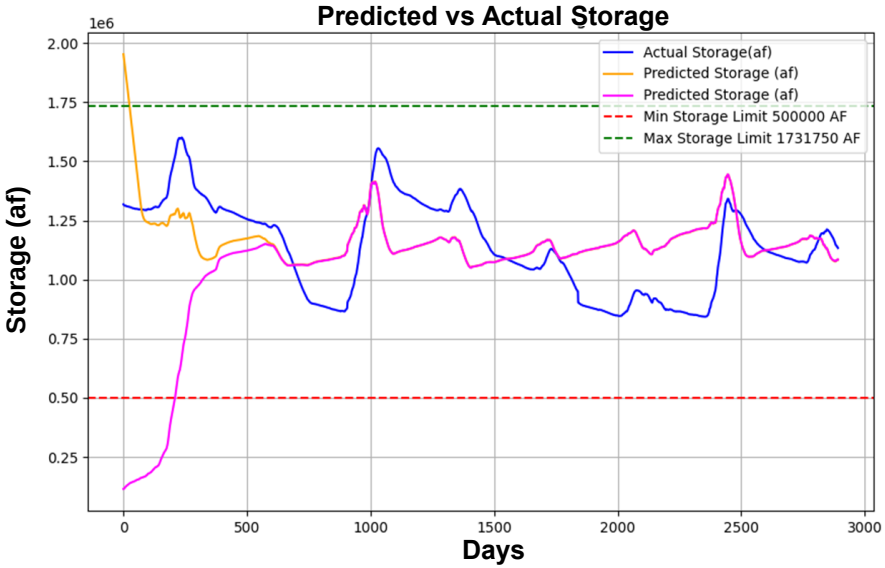
Smoothness Metric : 0.5308

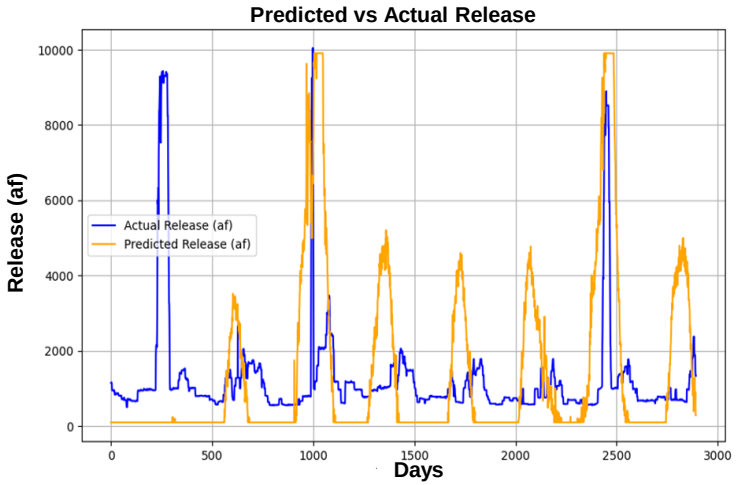


Smoothness Metric : 0.7762

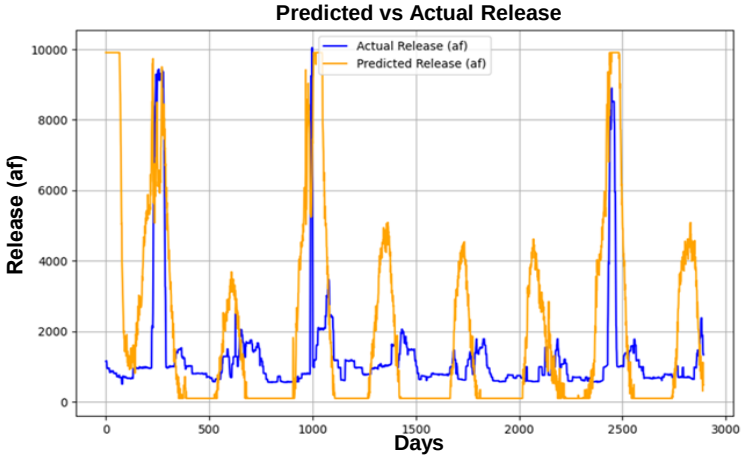
Stress Testing

Test 1: Initial storage 50 percent below/above the minimum/maximum storage limit





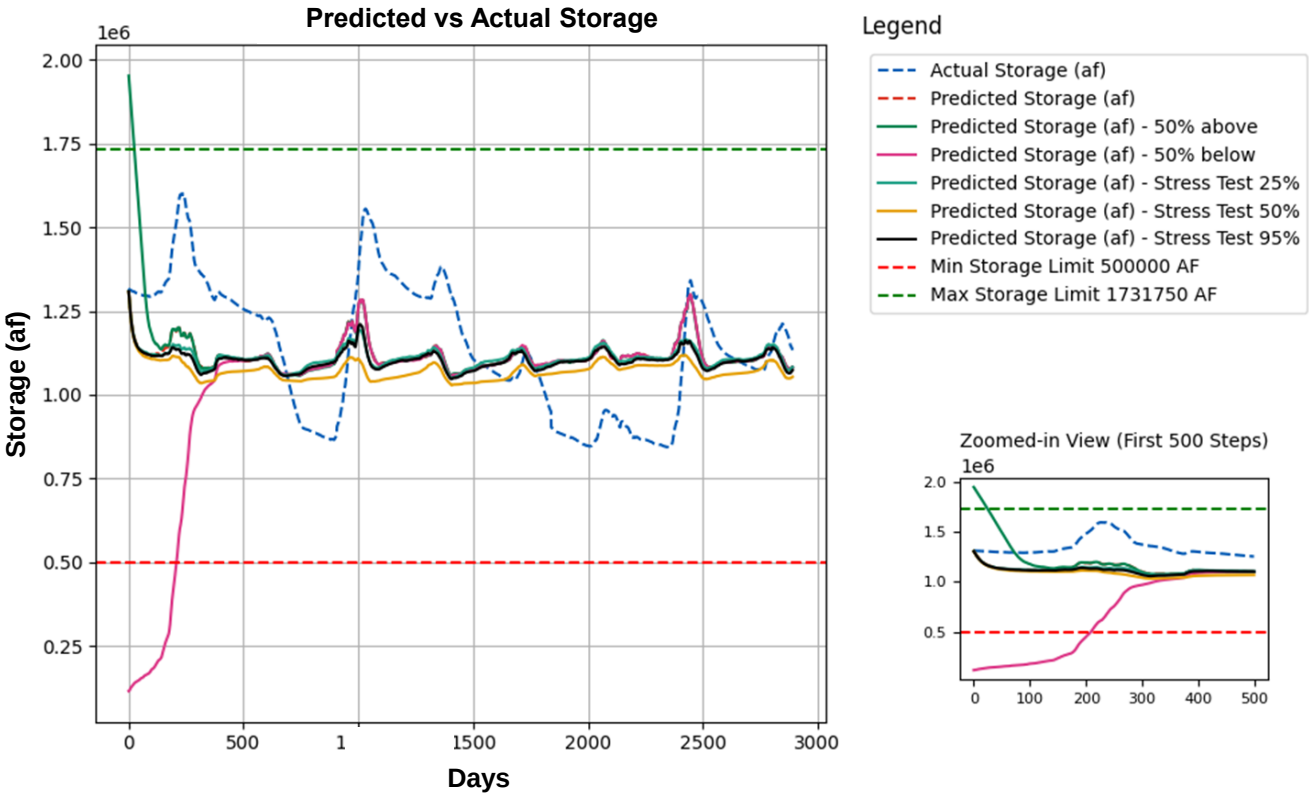
Release Pattern- 50% Below Min Storage



Release Pattern- 50% Above Max Storage

Stress Testing

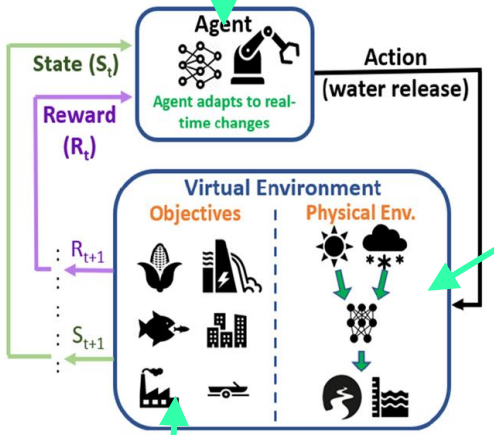
Test 2: Scaling inflows by 25%, 50%, and 90% (less available water into system)



Conclusions & Future Work

PPO Agent able to comply with basic constraints; flexible to multi-objectives, and resilient against out-of-sample stress tests

The framework shows promise, providing smooth and stable storage outcomes with flexibility to expand



Validated LSTM models for inflow and evaporation, allowing experimentation with alternative weather scenarios

Objectives' functional forms impact control in intuitive ways

⌛ Next Steps

- Using forecasts to guide decisionmaking
- Alternative weather scenarios (extreme drought and flooding)
- More complicated objective sets:
  - Honoring Endangered Species Act (release restrictions)
  - Hydropower generation
  - Water supply (municipal, agricultural, and water treaties.
  - Recreational constraints

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Thank You!